SEMESTER 7 · COMPUTER VISION PROJECT

CASE FILE
CXR · 5,856

GWCMC · PEDIATRIC CXR · AGES 1–5

Pediatric Pneumonia Classification from Chest X-rays

Kermany CXR · Guangzhou Women and Children's Medical Center

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LITERATURE SURVEY · BASELINE · HYPOTHESIS · 2018–2026

INTRODUCTION

What these slides cover.

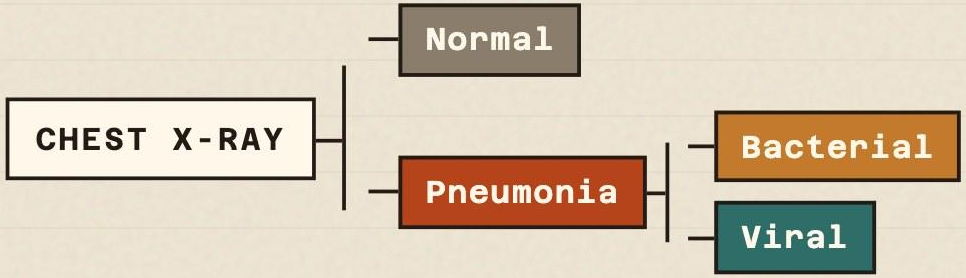
01 The data. One chest X-ray is normal or pneumonia. Pneumonia splits into bacterial and viral. The films are from Guangzhou. The children are 1 to 5 years old.

02 Ten papers, 2018 to 2026. Each paper uses its own split. These slides do not rank the papers on one accuracy number.

03 Our baseline. We reproduce XCCNet on the full Kermany set. The last slide states one hypothesis and one prediction. We state both before the next code change.

THE DATASET

Three labels on one film.



5,232
TRAIN IMAGES

624
TEST PATIENTS

1–5 yr
PEDIATRIC CXRS

Guangzhou
WOMEN & CHILDREN'S MEDICAL CENTER

Literature findings

01 Kermany et al., 2018 · *Cell*

02 Rajaraman et al., 2018 · *Appl. Sci.*

03 Stephen et al., 2019 · *J. Healthc. Eng.*

04 Rahman et al., 2020 · arXiv / *Appl. Sci.*

05 Salehi et al., 2021 · *Br. J. Radiol.*

06 Ayan, Karabulut & Ünver, 2021 · *Arab. J. Sci. Eng.*

07 Kundu et al., 2021 · *PLoS One*

08 XCCNet, 2024 · arXiv

09 Chauhan, Gupta & Doja, 2025 · LightPneumoNet

10 Transfer learning vs custom CNNs, 2026 · arXiv

01 / 10 · CELL 2018

Kermany et al.

INCEPTION V3 FROZEN | 2 BINARIES | OFFICIAL 624



MODEL / TASK

Inception V3, ImageNet, conv frozen, softmax only. Pneumonia vs normal + bacterial vs viral. No 3-class.

PREPROCESS

Quality screening. Remove unreadable images.

IMBALANCE

None. Live with 3:1 and 2:1.

TASK	ACC	SENS	SPEC	AUC
P vs N	92.8%	93.2%	90.1%	96.8%
B vs V	90.7%	88.6%	90.9%	94.0%

02 / 10 · APPL. SCI. 2018

Rajaraman et al.

CUSTOM VGG16 | CUT LAST LAYER · GAP + DENSE | 2 BINARIES + 3-CLASS | SAME 624



MODEL / TASK

Best: customized VGG16 (ImageNet). Cut the last layer, add GAP + dense. Also scratch sequential / residual / Inception. Both binaries + 3-class. CAM / LIME.

PREPROCESS

Atlas lung crop. Full film + ROI → 1024^2, mean-norm. No geo aug.

IMBALANCE

None. Keep noisy train images.

TASK (VGG16, CROP)	ACC	AUC	RECALL	F1 / SPEC
P vs N	96.2%	0.993	96.2%	F1 0.970
B vs V	93.6%	0.962	98.4%	Spec 86.0%
3-class	91.8%	0.939	90.0%	F1 0.910

03 / 10 · J. HEALTHC. ENG. 2019

Stephen et al.

SCRATCH CNN | 1 BINARY ONLY | RESHUFFLED SPLIT

N PNEUMONIA (B+V MERGED)



MODEL / TASK

4-block CNN from scratch. Sigmoid. Pneumonia vs normal only. Best input 200×200×3.

PREPROCESS

Resize, 1/255. Rot 40°, shift/shear/zoom 0.2, h-flip. No lung crop.

IMBALANCE

Heavy Keras aug. No class weights.

SPLIT	ACC	LOSS	AUC / SENS
Train	95.31%	0.129	—
Validation	93.73%	0.184	not reported

Rahman et al.

DENSENET201 WINS | 2 BINARIES + 3-CLASS | OWN ~200/CLASS TEST



MODEL / TASK

AlexNet, ResNet18, DenseNet201, SqueezeNet. MATLAB. All three tasks. 5-fold.

PREPROCESS

227² or 224². ImageNet norm. Rot 45°, 10% scale/shift. No crop.

IMBALANCE

Augment every class to 4,500.

DENSENET201	ACC	RECALL	AUC	F1
P vs N	98.0%	99%	0.98	0.981
B vs V	95.0%	96%	0.952	0.952
3-class	93.3%	93.2%	0.95	0.935

05 / 10 · BR. J. RADIOL. 2021

Salehi et al.

DENSENET121 | BINARY ONLY | HONEST / LOWER

N P · SAME KERMANY POOL, ~3:1



MODEL / TASK
VGG19, DenseNet121, Xception, ResNet50. Best DenseNet121. Pneumonia vs normal only.

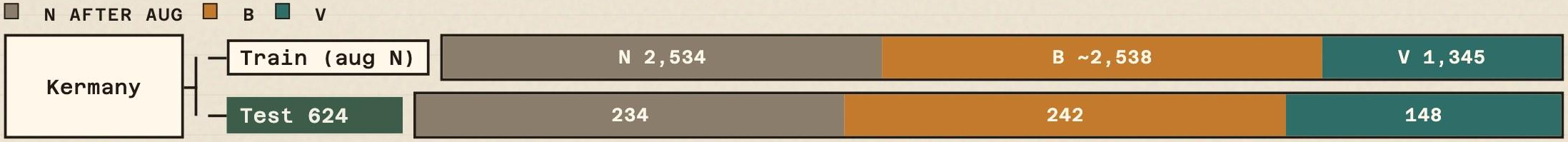
PREPROCESS
1/255, rot 15°, small shift/shear/zoom, flips, denoise, contrast, mean subtract.

IMBALANCE
SMOTE (rare in this field).

MODEL	ACC	AUC	SENS
DenseNet121	86.8%	0.86	>91%
Xception	86.0%	0.81	>91%
All four	≥83%	—	>91%

Ayan, Karabulut & Ünver

ENSEMBLE OF 3 | BINARY + 3-CLASS | ~624 TEST



MODEL / TASK

VGG / ResNet / Inception / Xception / MobileNet / SqueezeNet / PNet. Best = 3 models, not 7. Binary + 3-class.

PREPROCESS

TL, GAP, batch-norm, L2. Rot / zoom / flip on normals.

IMBALANCE

Normals 1,349 → 2,534 (near bacterial count).

TASK	ACC	SENS / REC	SPEC / F1
Binary ensemble	95.83%	97.76%	Spec 92.73% · AUC 95.21%
3-class ensemble	90.71%	B 97.93%	—
Viral class	—	77.03%	F1 82.31%

07 / 10 · PLOS ONE 2021

Kundu et al.

3-CNN ENSEMBLE | BINARY ONLY | KERMANY + RSNA



MODEL / TASK

Weighted-prob ensemble: GoogLeNet + ResNet-18 + DenseNet-121. Pneumonia vs normal.

PREPROCESS

Resize 224×224×3. No SMOTE.

IMBALANCE

Transfer learning only. No reweight.

SET	ACC	PREC	RECALL	F1 / AUC
Kermany	98.81%	98.82%	98.80%	F1 98.79% · AUC 98.35%
RSNA	86.95%	—	87.02%	drops ~12 pts

08 / 10 · ARXIV 2024

XCCNet

CNN + TRANSFORMER | BINARY | 80/10/10 NOT 624



MODEL / TASK
Dilated CNN (SFx) + contrastive transformer (CoTFx). Binary. Grad-CAM / Score-CAM / LIME.

PREPROCESS
CLAHE, lung seg (ResUNet++), rib suppression.

IMBALANCE
GAN/ADA: +2,624 synthetic normals (1,583 → 4,207).

SETTING	ACC	P / R / F1
No aug	96.29%	—
Traditional aug	97.18%	—
Full stack	99.76%	99.75%

09 / 10 · ARXIV 2025

LightPneumoNet · Chauhan et al.

388K-PARAM CNN | BINARY | INDEPENDENT TEST



MODEL / TASK

Scratch 4-block CNN, filters 16→128, 1.48 MB. Pneumonia vs normal.

PREPROCESS

224², RGB→gray, [0,1]. Rot 12°, zoom/shift/shear 0.15. No h-flip.

IMBALANCE

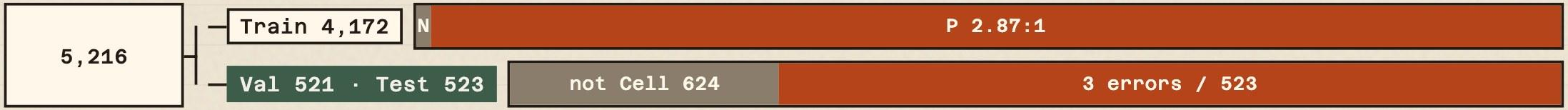
class_weight Normal 2.0, Pneumonia 1.2.

TASK	ACC	PREC	RECALL	F1
P vs N	94.2%	0.92	0.99	0.96

TL vs custom CNNs

RESNET50 FINE-TUNE | BINARY | 80/10/10 OF 5,216

N P · DISCARD 16-IMAGE VAL



MODEL / TASK

Scratch 4-block vs ResNet50 / DenseNet121 / EfficientNet-B0. Frozen vs last-2-block FT. Binary.

PREPROCESS

224², gray→3ch, ImageNet mean/std. Flip, ±10° rot, affine, jitter.

IMBALANCE

None. Report the 2.87:1 ratio.

RESNET50 FT	ACC	F1	AUC	SENS / SPEC
Headline	99.43%	99.61%	99.93%	99.48% / 99.26%
Note	FT beats frozen +5.48% avg. Ensemble added 0. Different exam than 624.			

DO NOT RANK ON ONE BOARD

Same set. Different exams.

#	PAPER	TASK	IMBALANCE	HEADLINE	SPLIT
1	Kermany	2 bin	none	92.8 / 90.7	624
2	Rajaraman	2 bin + 3	noisy train	96.2 / 93.6 / 91.8	624
3	Stephen	1 bin	heavy aug	93.7 val	custom
4	Rahman	2 bin + 3	4,500/cls	98 / 95 / 93.3	~200/cls
5	Salehi	1 bin	SMOTE	86.8	conservative
6	Ayan	bin + 3	aug N	95.8 / 90.7 · viral 77	~624
7	Kundu	1 bin	TL	98.8 → 87 RSNA	2 sets
8	XCCNet	1 bin	GAN	99.76	80/10/10
9	LightPneumo	1 bin	weights	94.2 / rec 0.99	held-out
10	2026 TL	1 bin	none	99.43	reshuffle

Hypothesis

We train XCCNet on all 5,856 Kermany films. The split is stratified 80/10/10. The run uses CLAHE and a lung mask. CLIP stays frozen. This run does not include the GAN. This run does not include rib suppression.

RESULT	ACC	PREC	RECALL	F1
Our test, 585 films	94.87%	94.81%	98.36%	96.55%
Paper, traditional aug	97.18%	94.52%	98.65%	96.54%
Paper, full stack	99.76%	99.75%	99.75%	99.75%

The model misses 7 pneumonia films and marks 23 normal films as pneumonia. Specificity is 85.4%. Sensitivity is 98.4%.

v1, dropped. "The gap comes from the GAN and rib suppression." We cannot build either from the paper. The paper's own precision and recall give 94.84% accuracy on our class mix. We got 94.87%.

v2 hypothesis. Replace cross-entropy with focal loss ($\gamma = 2$). Focal loss gives less weight to easy films, so training focuses on the hard normal films. **Prediction.** Specificity 85.4% → 88–91%. Sensitivity within 1 point. Accuracy +0.5 to 1.5. AUC unchanged (±0.005).

MILESTONE 3 · PREDICTIONS WRITTEN BEFORE EACH RUN

One change per row.

ROW	CHANGE	PREDICTED EFFECT	WHY
A	Baseline + AUC, specificity	AUC 0.97–0.99	30 errors in 585
A′	Threshold tuned on val	Spec +2–5, sens −1, same AUC	Moves FP into FN
B	Focal loss γ = 2	Spec 88–91%, AUC ±0.005	Hard normal films get the gradient
C	Lung mask off	Acc −0.5 to −2, CAM leaves lungs	Model can use markers and edges

If B beats A′, focal loss changed what the model learned. If B only matches A′, it just moved the threshold. γ is chosen on validation only. The test set is used once per run.

MILESTONE 4 · 585 TEST FILMS · MEAN ± STD OVER 3 SEEDS

Ablation results.

ROW	CHANGE	ACC	SENS	SPEC	AUC
A	Baseline	92.7 ± 0.6	95.3 ± 1.9	85.7 ± 3.3	0.978
A'	Threshold tuned on val	92.4 ± 0.7	93.3 ± 1.5	90.1 ± 2.2	0.978
B	Focal loss γ = 2	91.7 ± 0.4	96.3 ± 1.1	**79.1 ± 2.5**	0.975
C	Lung mask off	**95.8 ± 0.3**	**98.0 ± 0.4**	**89.9 ± 1.9**	**0.993**

ROW	WE PREDICTED	WE GOT
A'	Spec +2–5, sens −1	Spec +4.4, sens −2.0 ✓
B	Spec 88–91%, AUC ±0.005	**Spec 79.1% ✗ · AUC −0.003 ✓**
C	Acc −0.5 to −2	**Acc +3.1 ✗**

MILESTONE 5 · OUR HYPOTHESIS WAS WRONG

Focal loss only rescaled the scores.

01 Cross-entropy: 78% of films score below 0.1 or above 0.9. Focal loss: 35%. The median normal film moves from 0.08 to 0.35, so more normal films cross 0.5.

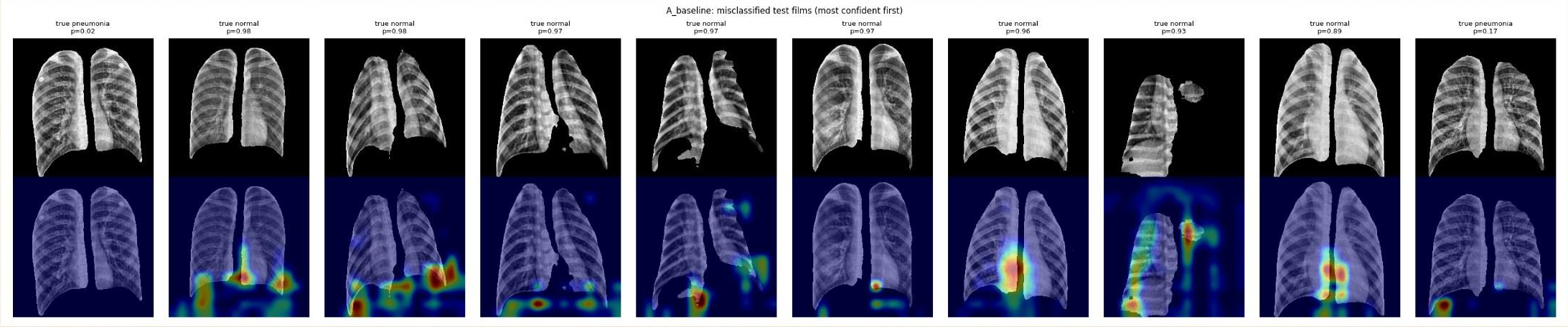
02 AUC does not change (0.978 vs 0.975). The ranking of films is the same.

03 With the threshold tuned on validation, both reach 90.1% specificity. B does not beat A′.

04 The risk we wrote down came true. The sampler already balanced the classes, so focal loss had nothing left to fix.

MILESTONE 5 · FAILURE ANALYSIS

The mask cuts off lung.



01 The mask model was trained on adult films. On child films it cuts off lobes. Normal films with the smallest masks: 23.4% error, against 14.4% for the rest.

02 Grad-CAM marks the mask edge and diaphragm, not lung texture.

03 Recall by subtype (row A): normal 85.7%, bacterial 96.5%, viral 93.4%. Viral is the weak pneumonia class, as in Ayan et al.

04 No mask scores higher, but the model may also use the "R" marker and film borders. One hospital cannot tell these apart.

REPRODUCTION GAP · LIMITATIONS

What the numbers can and cannot say.

SOURCE	ACCURACY
Paper, no GAN	97.18%
Paper's own precision + recall at our 427:158 mix	94.84%
Our first run	94.87%
Re-run, fixed split, 3 seeds	92.71 ± 0.55%

Between seeds, specificity moves by up to 6 points, so one seed is not enough. Limits: one hospital, pooled 80/10/10 split instead of the official 624-patient test, a stand-in lung mask, 3 seeds. **No clinical claim.**